Geophys. J. R. astr. Soc. (1972) **26,** 459–479

Palaeomagnetic Studies of Icelandic Lava Flows

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(Received 1971 July 8)

Summary

This paper describes a palaeomagnetic investigation of 107 igneous bodies, mostly basaltic lavas, of Pliocene and Pleistocene age in Iceland. A variety of conclusions has resulted from the study. Palaeomagnetic directions in lavas erupted in 1729, 1783 and 1875 do not support the postulated ' cycloid ' path for the Icelandic geomagnetic pole. Data are given for an application of geomagnetic reversal stratigraphy showing that Bering Strait opened more than 3 My ago, allowing Pacific boreal mollusks to enter Icelandic waters; these data also suggest that there were at least nine extensive glaciations since a time over 2 My ago. Lava flows erupted on Snaefellsnes Peninsula, western Iceland, contain large amounts of viscous remanent magnetization; partial demagnetization studies reveal a predominant reversed polarity for most of these lavas with one pronounced ' excursion ' of virtual geomagnetic poles to very low latitudes. Detailed studies of a dike cutting a lava flow show that the lava's remanence is unaffected beyond one dike width from the dike contacts. Brunhes epoch paleosecular variation is consistent with a model combining a changing nondipole field like the present one with a dipole wobble of about 11°.

1. Introduction

This report describes palaeomagnetic results derived from a collection of more than 800 samples obtained in 1964 from 107 igneous bodies (primarily basaltic lava flows) in Iceland. There were three primary objectives for this study: (1) to obtain further radiometric-polarity data to refine the then existent reversal time scale (Cox, Doell & Dalrymple 1964), (2) to use existing knowledge of the geomagnetic polarity reversal sequence to ' date ' the first appearance of boreal mollusks in Icelandic waters, and hence, the opening of Bering Strait (Einarsson, Hopkins & Doell 1967), and (3) to obtain a high northern latitude paleosecular variation value for the Brunhes normal polarity epoch (i.e. the last 0·7 My) to test geomagnetic secular variation models (Cox 1970).

Hospers (1953–1954) published the first extensive paleomagnetic study of Icelandic lavas, ranging in age from Tertiary to historic. Many other studies subsequently have appeared (Brynjólfsson 1957; Sigurgeirsson 1957; Einarsson 1957, 1962; Wensink 1964a, b; Dagley *et al.* 1967) so that the Icelandic volcanics have yielded a rather large amount of palaeomagnetic data. The present results duplicate some of the previous studies, supplement others, and also have yielded some new information.

The general locations of sample sites for this study are shown on Fig. 1 (detailed locations are given in the Appendix). The majority of the lava flows sampled were extruded on Snaefellsnes and Tjörnes peninsulas where sections with

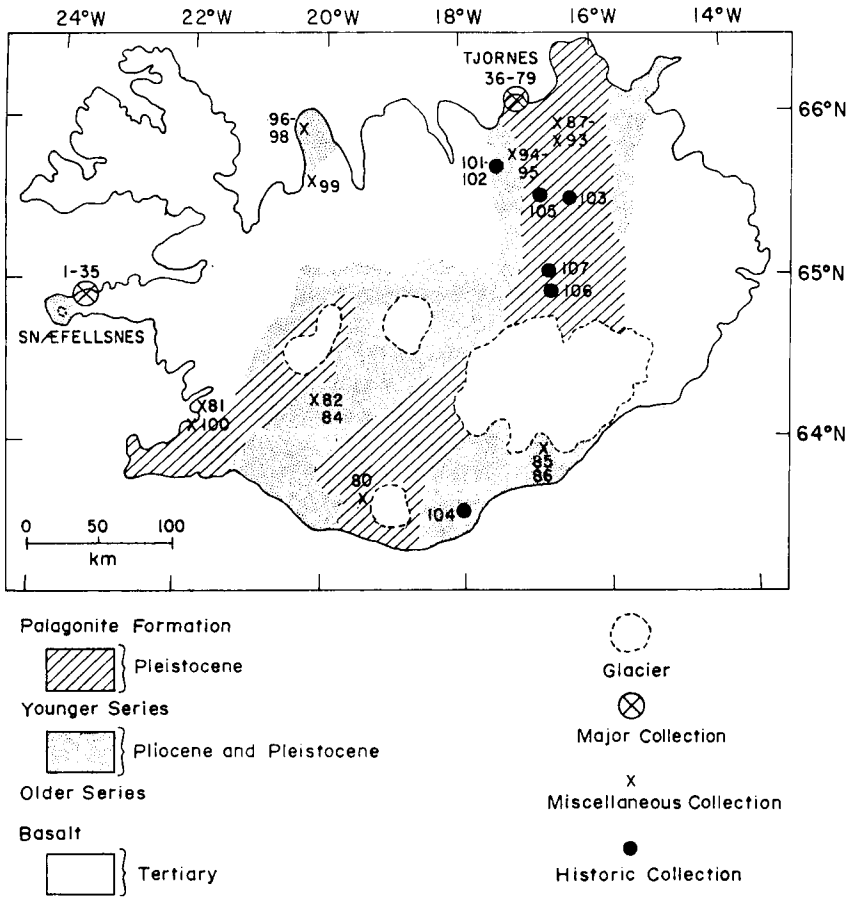


FIG. 1 Generalized geological map of Iceland (after Thorarinsson, Einarsson & Kjartansson 1959) showing location of sampling sites.

some stratigraphic and paleontologic control were available for realizing the objectives mentioned above. The collections from historically erupted and other miscellaneous lavas were obtained over most of the area of Pliocene and Pleistocene volcanism on Iceland. No effort was made to collect from the thick Tertiary sequences exposed to the east and west of the central zone of more recent volcanic activity.

2. Experimental procedures

Samples used in this study were 2·5-cm diameter cores about 5–15 cm long drilled *in situ* and oriented in geographical coordinates before removal from the outcrop (Doell & Cox 1967a). For the most part, eight cores spread over lateral distances of several tens of meters were drilled from each lava flow so as to get as representative a palaeomagnetic determination from the entire flow as possible. Details of the location of each lava flow sampled, along with maximum lateral separations of individual cores, are given in the Appendix.

In the laboratory, specimens 2·3 cm long were cut from each core, and subsequently measured for natural remanent magnetization (NRM) using spinner-type magnetometers (Doell & Cox 1965a). One or more specimens from each lava flow were later progressively demagnetized using alternating field (AF) demagnetization with peak field strengths of 25, 50, 100, 200, 400 and 800 Oe. In some cases, the rocks

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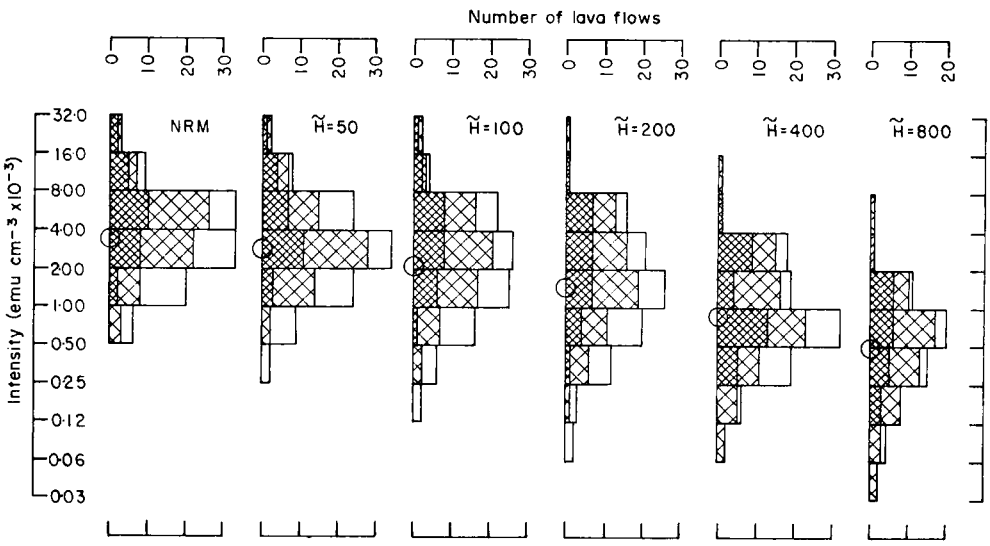


FIG. 2 Histograms of remanent magnetization remaining after partial alternating-field demagnetization at peak field strength $\tilde{H}$. Circles are mean values for each $\tilde{H}$. No pattern, lava flows from Snaefellsnes Peninsula; medium pattern, Tjörnes Peninsula lava flows; heavy pattern, historic and miscellaneous lava flows.

would not withstand treatment at the higher field strengths (Doell & Cox 1967b), and the demagnetization treatments were terminated below the highest values cited above. The results of these AF demagnetization treatments have been summarized in Fig. 2. As might be expected from this rather heterogeneous collection of lava flows, the stability against AF demagnetization varied rather widely. Moreover, some of the lavas possessed relatively large amounts of viscous remanent magnetization or isothermal remanent magnetization which was generally removed by the AF treatments. Thus, the values of NRM and RM remaining after AF treatment in low fields shown in Fig. 2 are not entirely due to thermoremanent magnetization (TRM). Nonetheless, it appears from the AF demagnetization studies that the majority of the lavas possess natural thermoremanent magnetizations of the order of 10^{-3} emu cm^{-3}.

Peak AF field strengths for magnetic cleaning of all samples from a given lava flow were chosen on the basis of the scatter in NRM directions and the behaviour of selected samples during progressive AF demagnetization. In several cases additional cleaning of a given lava flow was carried out if it appeared that this might decrease scatter. It has been our experience that inherent scatter in lava flows can generally be described by precision parameters, k (Fisher 1953) of several hundred (Doell & Cox 1963, 1965b; Doell 1969, 1970). (In this paper k_c will be used to denote the value of k after AF cleaning.) Thus, in the laboratory studies of the Icelandic lava flows, AF magnetic cleaning was often repeated using higher fields to obtain k_c values greater than 100 if possible. The palaeomagnetic data obtained from these studies are given in Tables 1, 2 and 3. In some instances not all samples collected from a given flow were used in the analysis, and this is indicated in the tables for individual lavas. Less than one-half per cent of the 833 samples collected were removed from analysis, and these exclusions were made in such a manner that the average direction cited for each lava in Tables 1, 2 and 3 is most probably that of the original TRM (Doell 1970). For one lava flow sampled (I36) no original TRM direction could be determined from the eight cores collected.

The directions cited in Tables 1, 2 and 3 are taken as the best estimates of the ancient geomagnetic field directions at the time and place of cooling of the lava. The precision with which these directions has been determined is given either by the angular standard

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Table 1

Historic and miscellaneous flow-average data

Unit No.	Number of samples †	A.F. cleaning (peak Oe)	Decl. (deg. E)	Incl. (deg. dn)	Length of resultant R	Precision parameter k_c	95% confidence α_{95} (deg.)	Angular s.d. (deg.)	VGP latitude (deg. N)	VGP longitude (deg. E)
I107	8 (8)	NRM	341·7	68·2	7·9834	422	2·7	3·9	73·3	206·4
I106	8 (8)	50	338·1	70·0	7·9272	96	5·7	8·3	74·5	218·4
I105	7 (8)	50	289·7	74·8	6·9905	634	2·4	3·2	60·0	279·2
I104	8 (8)	NRM	340·8	77·6	7·9810	368	2·9	4·2	81·5	279·2
I103	8 (8)	NRM	342·1	78·9	7·9847	456	2·6	3·8	82·5	283·7
I102	8 (8)	NRM	337·6	78·8	7·9852	473	2·5	3·7	81·0	279·3
I101	8 (8)	NRM	321·1	81·3	7·9631	190	4·0	5·9	74·9	297·2
I100	8 (8)	NRM	0·1	65·8	7·9619	184	4·1	6·0	73·9	157·9
I99	8 (8)	100	24·0	68·5	7·9660	206	3·9	5·6	71·5	107·0
I98	8 (8)	NRM	11·6	73·4	7·9741	270	3·4	4·9	81·4	116·0
I97	8 (8)	100	206·9	—79·2	7·9612	181	4·1	6·0	—79·4	220·6
I96	8 (8)	100	239·9	—75·1	7·9787	329	3·1	4·5	—64·5	230·2
I95	7 (8)	400	196·8	—51·7	6·9311	87	6·5	8·7	—55·0	317·5
I94	8 (8)	50	140·5	—79·8	7·9638	193	4·0	5·8	—74·9	107·1
I93	8 (8)	NRM	354·3	68·2	7·9940	1163	1·6	2·4	75·1	177·5
I92	8 (8)	25	43·1	83·7	7·9905	736	2·0	3·0	72·9	13·5
I91*	10 (10)	NRM	132·6	75·9	9·9910	1002	1·5	2·6	43·9	10·8
I90	8 (8)	NRM	352·5	68·3	7·9924	921	1·8	2·7	75·0	181·8
I89	8 (8)	100	42·6	71·7	7·9908	761	2·0	2·9	68·0	77·9
I88	8 (8)	NRM	315·5	69·8	7·9912	792	2·0	2·9	65·3	246·0
I87	8 (8)	NRM	45·4	80·3	7·9942	1200	1·6	2·3	73·1	36·3
I86	8 (8)	25	3·2	64·3	7·9679	218	3·8	5·5	72·0	155·9
I85	5 (8)	200	166·8	—47·5	4·9934	607	3·1	3·3	—53·7	2·9
I84	9 (9)	100	354·5	72·0	8·9960	1978	1·2	1·8	82·2	182·6
I83	8 (8)	NRM	353·7	64·0	7·9753	283	3·3	4·8	71·0	173·6
I82	8 (8)	25	286·6	69·1	7·9720	250	3·5	5·1	52·3	267·8
I81	8 (8)	50	149·0	—77·7	7·9841	442	2·6	3·9	—77·0	92·0
I80*	9 (9)	NRM	312·2	— 2·4	8·8175	44	7·9	12·3	16·2	211·0

†Number of sample cores used in analysis (number of sample cores collected).
*Not included in palaeosecular variation analyses.

Table 2

Tjörnes Peninsula flow-average data

Unit No.	Number of samples †	A.F. cleaning (peak Oe)	Decl. (deg. E)	Incl. (deg. dn)	Length of resultant R	Precision parameter k_c	95% confidence α_{95} (deg.)	Angular s.d. (deg.)	VGP latitude (deg. N)	VGP longitude (deg. E)
I79	8 (8)	NRM	7·3	80·7	7·9910	781	2·0	2·9	83·7	3·7
I78	8 (8)	NRM	355·2	81·4	7·9898	683	2·1	3·1	82·8	331·6
I77	6 (8)	50	37·9	78·2	5·9799	249	4·2	5·1	75·2	50·6
I76	8 (8)	50	27·7	81·4	7·9670	212	3·8	5·6	78·2	23·9
I75	8 (8)	NRM	353·6	83·4	7·9794	340	3·0	4·4	78·9	335·1
I74	8 (8)	50	0·5	68·9	7·9852	474	2·5	3·7	76·3	161·4
I73	8 (8)	NRM	357·5	77·7	7·9926	946	1·8	2·7	89·0	266·3
I72	8 (8)	NRM	23·2	75·5	7·9904	726	2·1	3·0	79·4	81·6
I71	8 (8)	NRM	184·8	—75·3	7·9939	1158	1·6	2·4	—85·6	312·6
I70	10 (10)	NRM	182·6	—70·9	9·9914	1051	1·5	2·5	—79·0	335·2
I69	8 (8)	NRM	190·7	—72·3	7·9876	563	2·3	3·4	—80·0	308·0
I68	8 (8)	NRM	195·8	—73·2	7·9912	794	2·0	2·9	—79·8	290·7
I67	7 (8)	50	212·7	—74·1	6·9840	374	3·1	4·2	—74·4	259·4

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Table 2 (*continued*)

I66	8 (8)	NRM	176·4	—76·7	7·9895	668	2·1	3·0	—87·9	32·2
I65	8 (8)	NRM	236·4	—72·9	7·9878	572	2·3	3·4	—63·7	242·8
I64	8 (8)	50	195·6	—74·7	7·9874	555	2·3	3·4	—81·6	280·8
I63	8 (8)	100	204·5	—76·1	7·9591	171	4·2	6·2	—79·4	255·4
I62	8 (8)	NRM	224·4	—72·3	7·9933	1041	1·7	2·5	—67·8	254·9
I61	9 (9)	100	199·0	—70·6	8·9574	188	3·8	5·9	—75·4	294·6
I60	7 (8)	NRM	215·2	—72·3	6·9697	198	4·3	5·7	—71·5	264·0
I59	8 (8)	NRM	15·8	65·5	7·9793	339	3·0	4·4	69·6	130·8
I58	4 (8)	50	90·5	—74·0	3·9826	172	7·0	6·1	—52·7	107·7
I57	8 (8)	100	4·9	68·2	7·9748	278	3·3	4·8	75·0	150·8
I56	8 (8)	NRM	6·7	67·0	7·9760	291	3·2	4·6	73·2	147·6
I55	8 (8)	25	338·3	63·2	7·9888	625	2·2	3·2	65·5	202·1
I54	8 (9)	25	356·8	64·6	7·9845	451	2·6	3·8	70·2	169·4
I53	8 (8)	50	27·8	72·3	7·9956	1607	1·4	2·5	74·4	93·6
I52	8 (8)	50	39·3	72·0	7·9850	466	2·6	3·8	69·6	80·6
I51	8 (8)	100	48·8	54·4	7·9816	381	2·8	4·1	47·9	95·8
I50	8 (8)	50	0·8	67·6	7·9908	760	2·0	2·9	74·3	160·8
I49	8 (8)	50	343·6	77·3	7·9827	404	2·8	4·0	83·3	256·4
I48	8 (8)	50	27·2	77·3	7·9852	473	2·5	3·7	79·0	62·4
I47	7 (8)	100	1·4	67·8	6·9757	247	3·8	5·2	74·6	159·3
I46	8 (8)	50	7·9	66·1	7·9817	382	2·8	4·1	71·8	145·8
I45	8 (8)	50	17·5	64·2	7·9938	1134	1·6	2·4	67·7	129·2
I44	8 (8)	50	21·4	65·0	7·9893	654	2·2	3·2	67·7	121·6
I43*	8 (8)	50	265·4	—56·0	7·9742	271	3·4	4·9	—34·8	239·9
I42*	8 (8)	50	263·8	—59·4	7·9875	558	2·3	3·4	—38·6	239·0
I41	8 (8)	100	139·7	—62·6	7·9636	192	4·0	5·8	—59·0	47·3
I40*	7 (8)	200	234·3	25·4	6·9442	108	5·8	7·8	— 1·0	290·5
I39*	7 (8)	50	252·8	—55·3	6·9572	140	5·1	6·8	—39·3	251·4
I38	8 (8)	NRM	325·1	84·2	7·9770	305	3·2	4·6	74·2	317·9
I37	6 (8)	NRM	215·0	—75·9	5·9802	253	4·2	5·1	—75·0	247·4
I36*	0 (8)	(no stable original remanent magnetization could be identified)								

†Number of sample cores used in analysis (number of sample cores collected).
*Not included in palaeosecular variation analyses.

Table 3

Snaefellsnes Peninsula flow-average data

Unit No.	Number of samples †	A.F. cleaning (peak Oe)	Decl. (deg. E)	Incl. (deg. dn)	Length of resultant R	Precision parameter k_c	95% confidence α_{95} (deg.)	Angular s.d. (deg.)	VGP latitude (deg. N)	VGP longitude (deg. E)
I35	6 (8)	NRM	167·2	—59·3	5·9955	1120	2·0	2·4	—64·0	359·5
I34	8 (8)	NRM	169·2	—59·3	7·9893	652	2·2	3·2	—63·8	355·7
I33	8 (8)	NRM	170·9	—59·2	7·9734	263	3·4	5·0	—64·5	352·9
I32*	3 (4)	NRM	201·1	—49·0	2·9989	1813	2·9	1·9	—52·6	305·7
I31	8 (8)	100	202·2	—51·6	7·9943	1238	1·6	2·3	—54·5	303·2
I30	6 (6)	100	188·2	—75·1	5·9934	756	2·4	2·9	—85·3	281·1
I29	6 (6)	50	184·9	—74·0	5·9839	311	3·8	4·6	—84·8	308·8
I28	6 (6)	200	189·4	—72·5	5·9883	427	3·2	3·9	—81·5	300·3
I27	15 (15)	100	176·7	—62·9	14·8830	120	3·5	7·4	—69·3	343·3
I26‡	8 (39)	50	224·2	—80·7	7·9922	901	1·8	2·7	—72·8	203·9
I25	6 (8)	100	211·0	—68·0	5·9897	486	3·0	3·7	—68·9	272·7
I24*	3 (5)	200	123·8	—57·3	2·9839	124	11·1	7·3	—47·9	54·7
I23*	4 (5)	100	111·0	—50·1	3·9906	320	5·1	4·5	—36·5	62·1
I22*	5 (5)	200	55·9	—54·7	4·9929	561	3·2	3·4	—19·2	110·8
I21*	4 (6)	200	0·2	—74·6	3·9978	1357	2·5	2·2	—36·1	156·5
I20*	5 (6)	100	342·2	—70·6	4·9819	221	5·1	5·4	—30·6	168·4
I19	5 (5)	100	254·3	—81·7	4·9807	207	5·3	5·6	—64·4	195·2

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Table 3 (*continued*)

I18	5 (5)	100	297·9	—57·4	4·9959	983	2·4	2·6	—57·4	177·2
I17	3 (5)	200	288·2	—78·5	2·9946	373	6·4	4·2	—52·1	192·2
I16	4 (5)	100	193·4	—62·6	3·9904	314	5·2	4·6	—67·7	310·5
I15	4 (5)	100	201·8	—60·5	3·9807	156	7·4	6·5	—63·4	298·1
I14	5 (5)	100	190·9	—59·6	4·9905	419	3·7	4·0	—64·7	316·8
I13	5 (5)	100	186·4	—60·3	4·9962	1046	2·4	2·5	—66·0	324·7
I12	4 (4)	100	185·5	—56·5	3·9964	827	3·2	2·8	—62·0	327·2
I11	5 (5)	100	199·1	—62·3	4·9563	92	8·0	8·5	—66·1	300·7
I10	5 (5)	100	194·3	—62·1	4·9908	436	3·7	3·9	—67·0	309·3
I9	4 (5)	NRM	182·0	—60·2	3·9952	622	3·7	3·2	—66·2	332·8
I8	5 (5)	200	184·1	—61·3	4·9960	993	2·4	2·6	—61·3	184·1
I7	6 (8)	100	216·0	—71·2	5·9868	380	3·4	4·2	—70·1	256·4
I6	9 (9)	50	184·4	—64·1	8·9727	293	3·0	4·7	—70·8	327·1
I5	8 (8)	100	2·8	83·8	7·9572	164	4·3	6·3	77·2	339·2
I4	9 (9)	50	25·1	74·6	8·9610	205	3·6	5·7	78·1	73·5
I3	8 (8)	100	23·1	73·8	7·9641	195	4·0	5·8	78·2	81·4
I2	4 (8)	100	37·4	76·1	3·9953	638	3·6	3·2	74·0	53·9
I1	7 (7)	100	29·1	67·3	6·9720	215	4·1	5·5	68·8	96·8

†Number of sample cores used in analysis (number of sample cores collected).
*Not included in palaeosecular variation analyses.
‡See text for details of effects of cross-cutting dike on this flow.

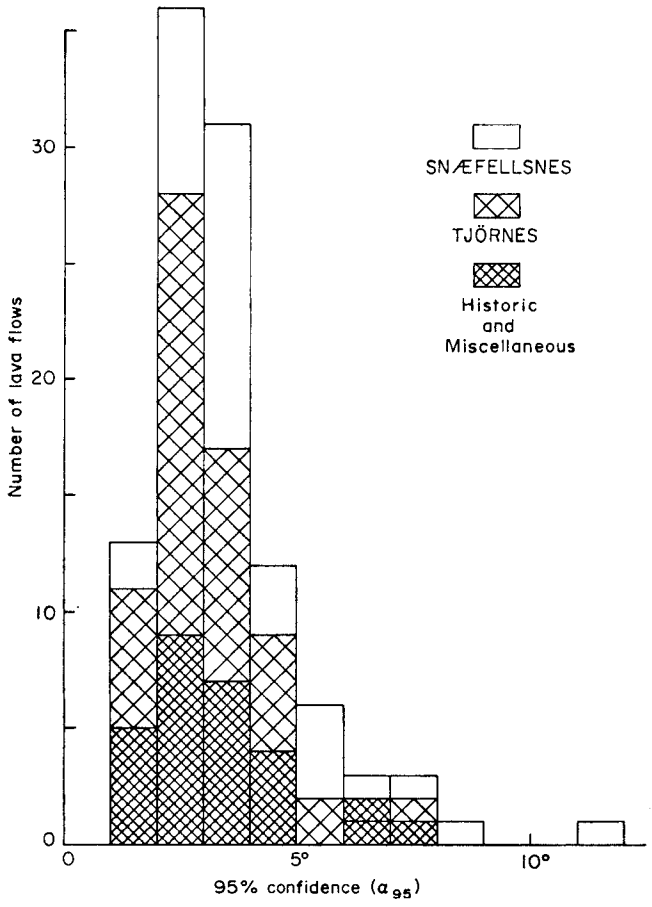


FIG. 3. Histogram of 95-per-cent-confidence cone semiangles (α_{95}).

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deviation or the 95-per-cent-confidence semiangle α_{95} (Fisher 1953). The overall precision of the data is summarized in Fig. 3 by means of a histogram of the flow-average α_{95} values. At the 95-per-cent-confidence level, most of the directions are defined within a cone of 5–6° radius, which is relatively small with respect to the average angles between flow mean directions.

3. Discussion of results

In addition to the reversal time scale attempts and to the palaeomagnetic study of Tjörnes Peninsula lava flows that was undertaken primarily for stratigraphic correlation purposes (Einarsson *et al.* 1967), several other results were also obtained from these Icelandic collections. These are based on (1) a few collections from historically erupted lava flows made to supplement the earlier work of Brynjólfsson (1957) and Hospers (1953–1954); (2) some resampling of sections on Snaefellsnes Peninsula, originally measured by Hospers (1953–1954); (3) detailed sampling of a lava flow near a cross-cutting dike; and (4) additional collections of Brunhes polarity epoch lavas to obtain values for paleosecular variation in Iceland.

3.1 *Lava flows erupted in historic times*

Collections were made from lava flows erupted in 1729, 1783, 1875, 1921, and 1961 (unit numbers I 103 to I 107 (Table 1)). Brynjólfsson (1957) had previously

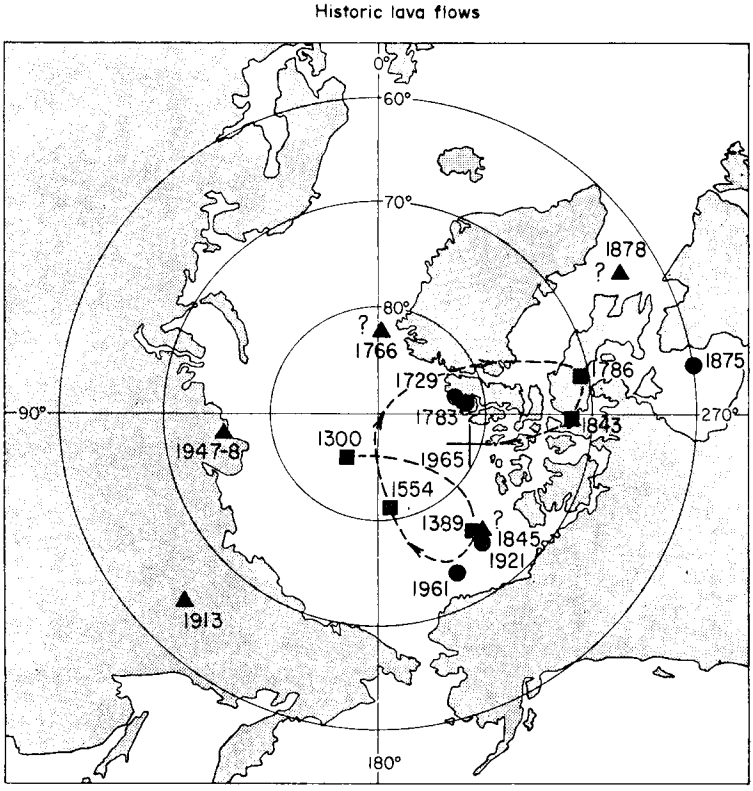


FIG. 4. Virtual geomagnetic poles determined from lava flows erupted in historic times. Squares and dashed curve from Brynjólfsson (1957); triangles from Hospers (1953–1954); circles from present studies. Present VGP (cross) is calculated from the International Geomagnetic Reference Field for 1965 (IAGA Commission 1969). Portion of Lambert equal-area projection.

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reported on palaeomagnetic studies of lavas with ages of 1300, 1389, 1554, 1786, and 1843 A.D., and he drew a virtual polar path from these data that suggested a ' cycloidal ' motion to Kawai, Hirooka & Sasajima (1965). The present collections were made in an attempt to add more detail to this path. The VGPs from these data are shown with those previously reported by Brynjólfsson along with data from Hospers (1953–1954) in Fig. 4. It can be seen that the new data do not add much detail (or confirmation) to the path suggested by Brynjólfsson.

It must also be noted that the data for the flows erupted in 1961 and 1921 yield VGPs quite removed from the VGP for Iceland during this century (which has had a position close to that shown for 1965). There are three possible reasons for this discrepancy that warrant discussion. First, over the distance spanned by the samples (46 m for the 1961 lava and 52 m for the 1921 lava) there may have existed uniform anomalies over these distances that deflected the local field some 10° away from the regional field direction. Both of these lavas were sampled in or near the summit crater of Askja Volcano where such anomalies might be expected (Decker 1963). During the procedure used for geographically orienting samples, the angle between the regional magnetic declination and the local declination about 10–20 cm above each sample core was measured (Doell & Cox 1963); for the eight samples from the 1961 lava this mean declination deviation is only 3·1°, but for the 1921 lava the mean of the eight measured declination deviations is 15·2°. However, as may be seen from the location of the VGPs with respect to Iceland in Fig. 4, the displacements away from the true VGP are caused by shallowing of inclination, rather than changes in declination, and present inclination anomalies are not known. Local anomalies over the distances spanned by the samples, if they existed, were therefore quite different before cooling of these lavas than at present. A second possibility is that there was tilting of the surface and edges (which were the only portions available for sampling) of these two flows after the magnetization had been acquired. This has not been observed before (Doell & Cox 1963); moreover, the lavas are no more than 1–2 m thick where sampled, and it is unlikely that a portion of the flow 50 m long could have been uniformly tilted 10° without introducing considerably more core-to-core scatter in directions than was observed (Table 1). A third possibility might be that the RM acquired by these lavas was not parallel to the ambient field at the time they cooled. Again, this would be contrary to previous paleomagnetic results from lavas erupted during times when the direction of the regional geomagnetic field was known from direct measurements (Doell & Cox 1963).

The most likely cause of the discrepancies for the 1921 and 1961 VGPs therefore appears to be due to the ' local ' anomalies associated with the summit region of Askja Volcano. The results of Hospers (1953–1954) for the 1913 and 1947–48 lavas also yield VGPs far from the expected positions and this may likewise be due to close proximity to a volcano summit—in this latter case, that of Mount Hekla. (Mean directions, from which the VGPs attributed to Hospers were determined, were calculated from data scaled from Fig. 2 of Hospers (1953–1954) using Fisher's (1953) statistical methods. Ninety-five per cent confidence limits for the 1913 and 1947–48 directions were less than 10° whereas the others exceed 18° and are marked with a question mark in Fig. 4.)

The other historic results reported here from lavas erupted in 1729, 1783 and 1875, were all collected on the flanks of volcanoes, well removed from possible summit anomalies. They therefore should not be in error as are the summit lavas of 1921 and 1961. The VGPs for these lavas are in the same general region as reported by Brynjólfsson for 1786 and 1843, but they fall rather farther from the appropriate positions on the ' cycloid ' path than could be attributed to experimental error. It seems that virtual polar movement in Iceland during these times may have been different or more complicated than previously suggested (Brynjólfsson 1957; Kawai *et al.* 1965). However, more data would be desirable before definitive statements

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could be made concerning geomagnetic field behaviour in Iceland during the last several centuries.

3.2 *Tjörnes Peninsula lava flows*

The polarity and stratigraphic relationships of the lava flows studied on Tjörnes Peninsula have been discussed previously (Einarsson *et al.* 1967) and are summarized in Fig. 5. These data provided evidence that the first appearance of Pacific boreal mollusks in Icelandic waters took place more than 3 My ago, suggesting that this was the first opening of Bering Strait. The data also were used to show that there were at least nine extensive glaciations, separated by lengthy non-glacial intervals, that probably commenced over 2 My ago. Wensink (1964a) has reported similar evidence from other sequences in eastern Iceland.

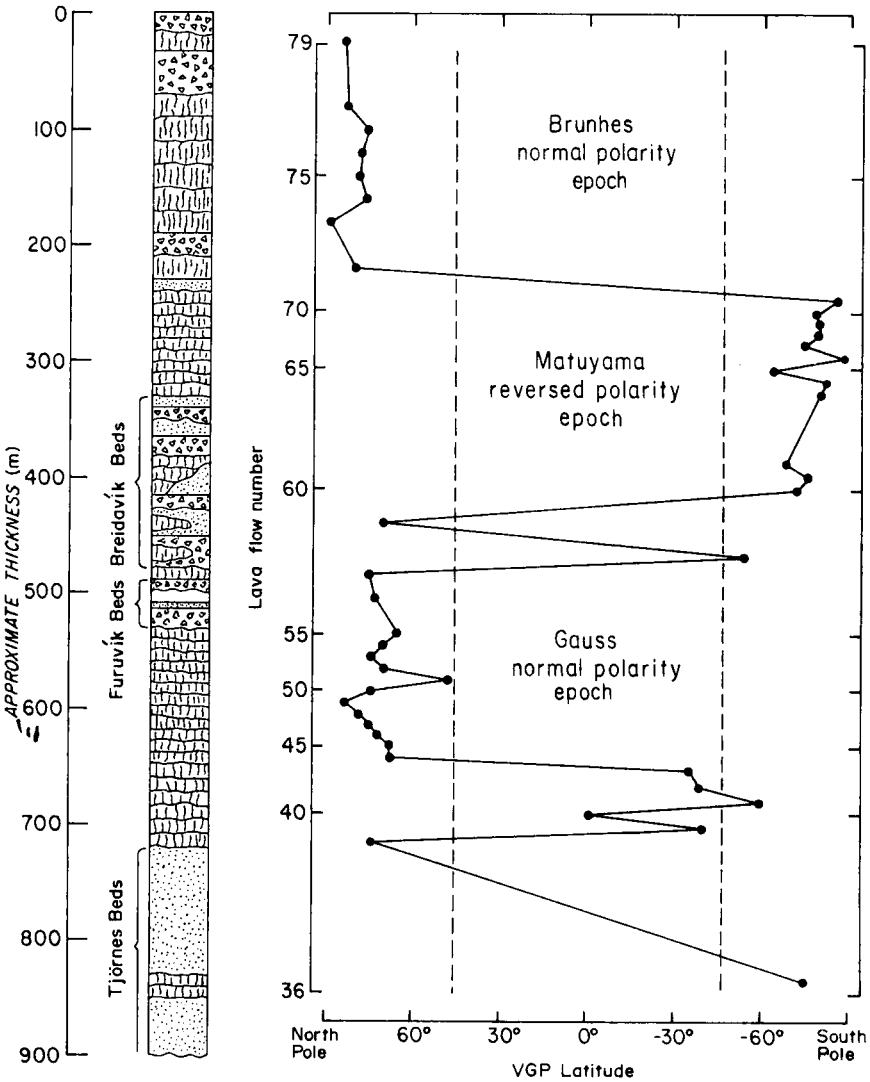


FIG. 5. Virtual geomagnetic pole latitude plotted against approximate stratigraphic position for Tjörnes Peninsula lava flows. Schematic section adopted from Einarsson, Hopkins & Doell (1967). Dashed lines at ±46° latitude. The blocky pattern represents lava flows, the dotted pattern represents sediments, and the triangular pattern represents tillites.

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Here, we wish only to give details of the palaeomagnetic data (Table 2) and to include them in the overall analyses for palaeosecular variation given below. A few other comments are also appropriate. From the Appendix, it may be noted that it was not possible to distinguish, geologically, whether the following pairs of sampling sites consisted of different lava flows or the same lava flow: I42–I43, I46–I47, I48–I49 or I50, I52–I53, I64–I65, and I72–I73. From the data given in Table 2 it is quite likely that sites I42 and I43 sampled the same lava since both produce nearly the same rather unusual direction. I46 and I47 could be the same, but need not be. All of the other site pairs consist of distinct lavas since the corresponding directions are significantly different from each other (Table 2).

The five lava flows I39 through I43 are taken to represent the Mammoth event within the Gauss normal polarity epoch (Einarsson *et al.* 1967). From Fig. 5 it is clear, however, that most of the VGPs calculated from these data fall in rather low latitudes—I40 being essentially on the equator—and are best described as 'intermediate' (with the exception of I41). There is no apparent relationship in directions going from one flow to another: the flows I39 and I42–I43 have similar directions,but the intervening flows have directions far removed from these and from each other. As mentioned above, I42 and I43 are probably collections from the same lava, but no reason why I39 has such a similar unusual direction is apparent.

3.3 *Snaefellsnes Peninsula lava flows*

As mentioned earlier, the primary purpose of recollecting the sites visited by Hospers (1953–1954) on Snaefellsnes Peninsula was to add data to the geomagnetic polarity epoch time scale; this was not realized because no lavas from this area were found to be suitable for age determinations (Dalrymple & Lanphere 1969). Some of the palaeomagnetic results are of interest by themselves, however.

As a preliminary to collecting in this area, a number of lava flows on Stöd were tested for polarity using field methods (Doell & Cox 1967c). The results of these studies are indicated on the map on Fig. 6. Like Hospers (Fig. 2 of Hospers (1953–

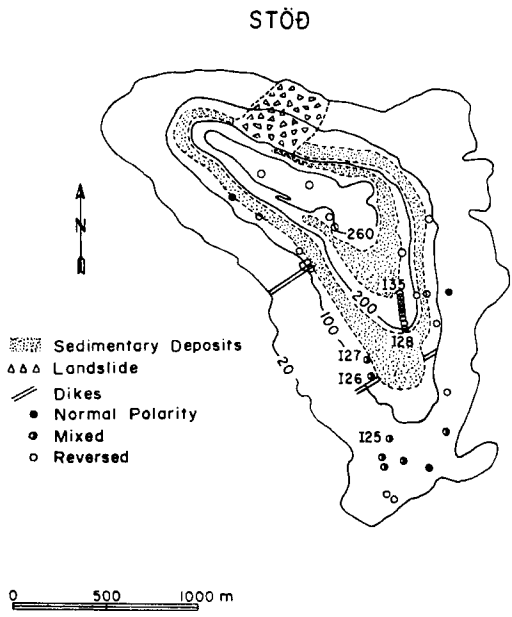


FIG. 6. Map of Stöd, Snaefellsnes Peninsula, showing locations and results of field polarity determinations. Lava flows also sampled for later laboratory studies are numbered I25 to I35.

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1954)), we found normal polarity below the lower sediment exposed on Stöd, whereas all the lavas younger than the sediments were reversely magnetized. Eight lavas above the sediments and three lavas below were also cored for later more detailed laboratory studies. Upon partial demagnetization, all of the older flows became reversely magnetized (Table 3), indicating that the normal NRM directions measured in the field were due to a viscous remanent magnetization (VRM) larger than the original reversed TRM. We do not know, of course, whether all of the lavas exposed below the sediments on Stöd have original reversed TRM, but it should be assumed that they do until AF demagnetization studies can be performed on those indicating a normal total NRM.

In the section exposed at Búlandsgil, Hospers (1953–1954) found a section of normally magnetized lavas overlying a reversely magnetized section, with the change in polarity occurring at an elevation of about 65 m. We sampled 17 superposed lava flows at this locality spanning the reversed-to-normal transition of Hospers. Field measurements of NRM in our samples generally agreed with Hospers' results, but upon partial demagnetization we found that all of the lavas possessing normal NRM had either reversed or ' intermediate ' original TRM. The data are given in Table 3, and a plot of the corresponding VGPs is shown in Fig. 7. During extrusion of

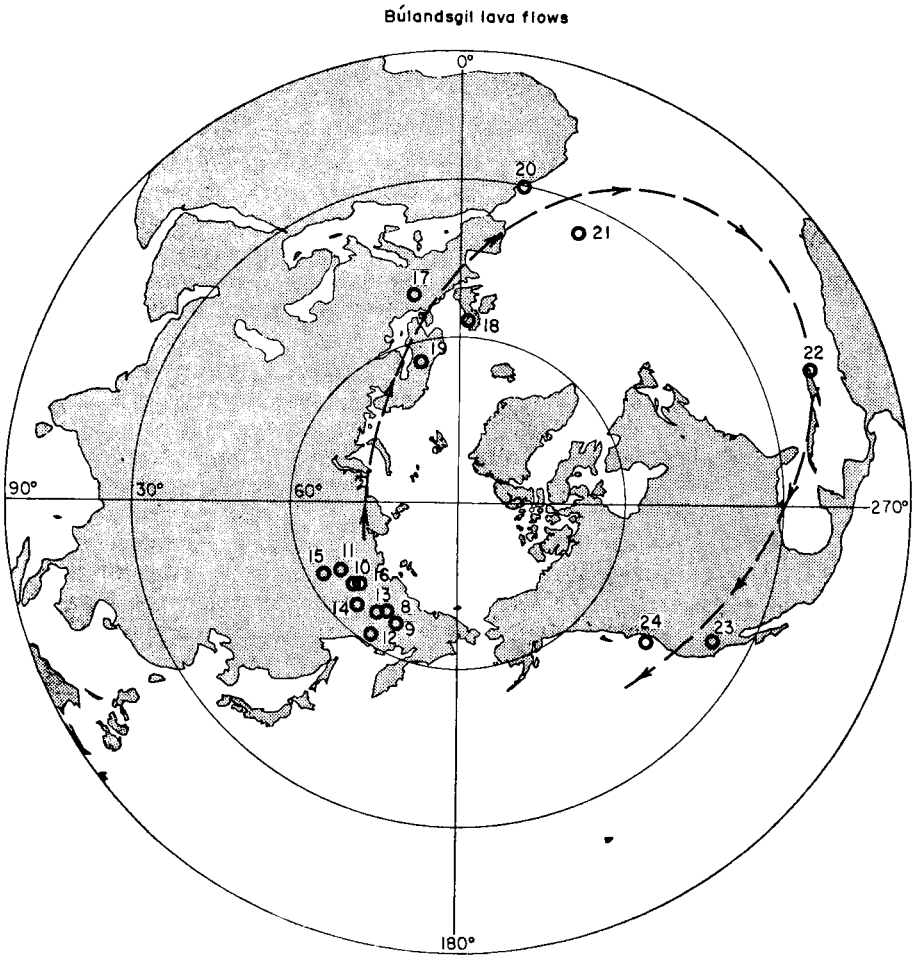


FIG. 7. VGPs calculated from cleaned magnetic directions of superposed lavas at Búlandsgil, Snaefellsnes Peninsula. Lavas are numbered from oldest, I8, to youngest, I24. Lambert equal-area projection.

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flows I8 through I16, there was very little apparent change in the geomagnetic field direction (suggesting rapid accumulation of these lavas). However, during extrusion of the following eight lavas, the field undertook a marked excursion, with corresponding VGPs reaching very low latitudes. This might be due to a large changing nondipole component (large relative to the dipole field) or to marked changes in the orientation of the main geomagnetic dipole. From these data alone it is not possible to choose which of these two alternatives is most likely. As may be noted in Table 3, different amounts of AF field were needed to magnetically clean the Búlandsgil samples, making it difficult to compare intensities of original TRM from one lava to another. However, there may be a difference in mean intensities of magnetization between the group I8 through I16 and those with low latitude VGPs. The mean intensity for the seven lavas demagnetized at 100 Oersteds in the older group is $1{\cdot}63 \pm 0{\cdot}38$ (s.d.) emu cm^{-3}, whereas that for four similarly demagnetized flows in the younger group is $1{\cdot}07 \pm 0{\cdot}03$ emu cm^3. If this difference implies different geomagnetic paleointensities, then it may be inferred that the 'excursion' is due to a lessening of the main dipole strength and hence a larger non-dipole to dipole moment ratio during the interval in question.

It is worth noting that Hospers (1953–1954) carried out his studies without the aid of partial demagnetization treatments to magnetically clean the samples, and thus he could not have been aware of the large amounts of VRM carried by some of the Snaefellsnes lavas. Magnetic polarity measured in the field has also been used for magnetic mapping and stratigraphic correlation in Iceland (Einarsson 1957, 1962). It can be expected that the bulk of these studies have yielded the correct polarity. On the other hand, if there is any doubt about a given study, such as those on Snaefellsnes, their reliability should be checked by partial demagnetization.

Hospers (1953–1954) and Einarsson (1957, 1962) generally separate a lower Tertiary sequence of lavas and an upper Quaternary sequence on Snaefellsnes at the sediments here informally referred to as the Búlandsgil sediments. There seems little doubt that the reversed lavas above the Búlandsgil beds were extruded during the Matuyama reversed polarity epoch. However, in view of the AF demagnetization results on the lower sequence, which changed an apparent normal polarity to a reversed polarity, it may be that at least some of the lavas below the Búlandsgil beds were also extruded during the Matuyama epoch. It should be pointed out that an unconformity below the Búlandsgil sediments has been recognized by Einarsson (1962) and assumed to indicate a substantial time break (the normal lavas farther west at Máfalid and also below the Búlandsgil sediments must clearly precede the Matuyama epoch, for example). However, there are also unconformities and sediment beds higher in the sequence, so that such features may not everywhere indicate substantial time breaks in this tectonically active area.

For the palaeomagnetic and variance analyses discussed later, Hospers' and Einarsson's interpretation is followed, and all lavas below the Búlandsgil sediments are assumed to be pre-Matuyama epoch, even though this may not be entirely correct.

3.4 *Effects of reheating by dikes*

When field polarity measurements were made at site I26 on Stöd, it appeared that a reversely magnetized dike system had cut across normally magnetized lava flows. This seemed to be an ideal situation in which to test the effects of reheating by dikes on an original TRM, and a large number of samples was drilled from two branches of a dike cutting a single lava flow and from the flow. As noted above, it turned out that the normal NRM of the Stöd lavas was due to a predominant VRM and that the original TRM is reversed. Thus the situation is not as ideal as first appeared; however, there seems to have been a difference of about 20° in the field direction between the times of cooling of the lava flow and dikes, and some effects were noted. Flow samples more distant than 1 m from the dike have RM directions with declina-

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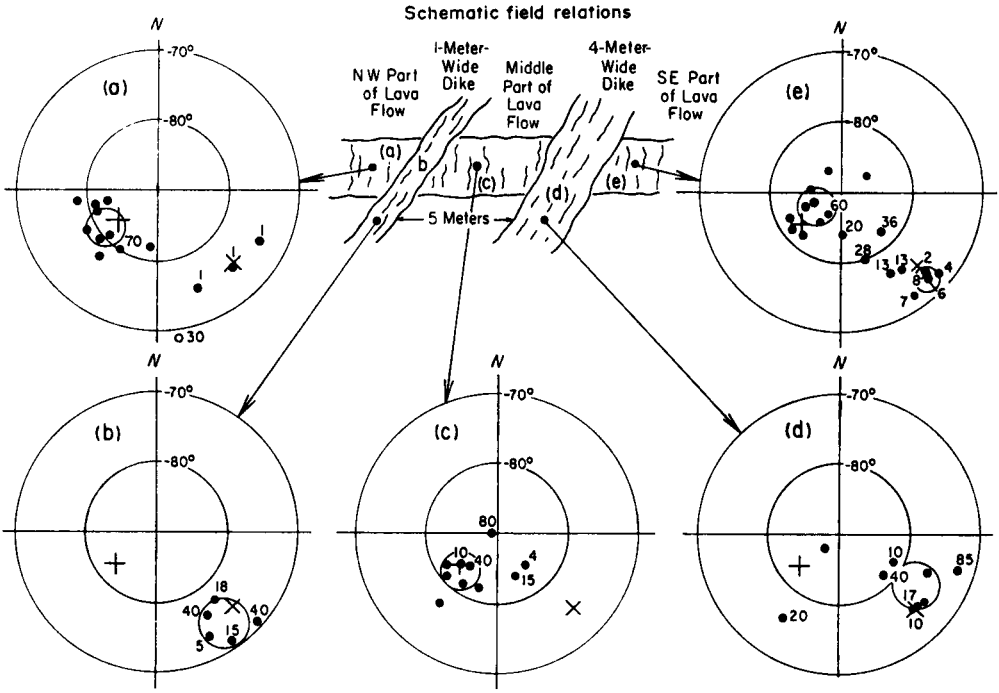


FIG. 8. Directions of remanent magnetization of individual samples from a lava flow and two branches of a cross-cutting dike near their mutual contacts. Numbers near samples are distances, in centimeters, from nearest dike-flow contact. Unnumbered samples are more than 100 cm from contacts. X = mean dike direction; + = mean flow direction; circles = 95-per-cent-confidence limits (see text for details). Portions of Lambert equal-area projections.

tions to the southwest and negative inclinations near 80° to 90°. Declinations of dike samples are typically to the southeast with negative inclinations near 75°. The overall mean direction for the lava flow (+ in Fig. 8) was determined from all samples one or more metres distant from the dike contacts, and that for the dike (X) from all sample directions in diagrams (b) and (d) of Fig. 8 except for the two samples in the south-western quadrant of (d) which were excluded from the determination for the mean.

For the left (NW) portion of the flow (a), 10 samples 70 cm or more from the contact have a mean direction not significantly different from the overall flow mean direction. Three samples essentially adjacent to the contact have directions very close to the dike mean direction, and thus appear to have been reheated above their Curie points. A study of diagrams (c) and (e) of Fig. 8 suggests that samples more than 40 to 60 cm from the contacts have directions that are little affected by reheating from the dike, and that only within about 10 cm of the dike is the lava flow completely remagnetized by reheating. In diagram (c) the 95 per cent confidence circle was calculated excluding samples within 15 cm of the contact; in diagram (e) the ' unaffected ' 95 per cent confidence circle was calculated excluding samples within 36 cm of the dike, and the ' completely affected ' circle was based on samples within 8 cm of the contact.

These kinds of effects could certainly be more precisely determined in situations where the lava flow and dike have opposing polarities. However, this study does indicate that the effects of dike reheating are not great. It is safe to conclude that RM directions of lava flow samples collected more than one dike width away from dike contacts should be unaffected by reheating due to dike intrusion.

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Table 4

Statistical analyses of flow-average data

Data analysed	Lat.	Long.	α_{95}	N	R	K_T	S_T
Brunhes-age VGPs	87·7° N.	137·7° W.	5·7°	28	26·8810	24	16·6°
Matuyama-age VGPs	79·7° S.	55·8° W.	6·3°	28	26·6333	20	18·4°
Pre-Matuyama-age VGPs*	78·4° N.	121·9° E.	5·4°	37	35·1874	20	18·4°
	Incl.	Decl.	α_{95}	N	R	k_T	s_T
Brunhes-age RM directions	+76·1°	5·0° W.	3·3°	28	27·6170	70	9·7°
Matuyama-age RM directions	−71·1°	169·9° W.	4·2°	28	27·3814	44	12·3°
Pre-Matuyama-age RM directions*	+70·2°	12·3° E.	3·3°	37	36·2968	51	11·4°

Notes: α_{95} = 95% confidence angle about means.
N = Number of lava flows (sites).
R = Vector sum of N unit vectors.
K_T, k_T = Total precision parameter, VGPs and RM directions, respectively.
S_T, s_T = Total angular standard deviation about means, VGPs and RM directions, respectively.
* = All reversed data converted to normal north-seeking data before analysis.

3.5 *Paleosecular variation*

The general paleomagnetic results obtained from the present collection of Icelandic lava flows are given in Table 4. The analyses have been made, using Fisher's (1953) methods, on RM directions and corresponding VGPs for Brunhes age lavas, Matuyama age lavas, and pre-Matuyama age lavas. The 95 per cent confidence limit for the mean of the 28 Brunhes age VGPs includes the present geographic pole, but those for the older two groups do not. However, the Matuyama and pre-Matuyama mean VGPs fall on opposite sides of the geographic pole so that if all data are combined, the resulting overall mean VGP is very close to the axis of rotation. For reasons discussed below, all flow-average data yielding VGPs with latitudes less than 46° have been excluded from these analyses.

Of more interest here is the angular dispersion in these data groups expressed by the values in the last two columns of Table 4. Before these can be interpreted directly as measures of ancient geomagnetic secular variation, it is necessary to remove the contributions due to experimental errors and to possible ancient local anomalies. These further analyses have been summarized in Table 5—again for both RM directions and VGPs for the three different age groups. Standard analysis of variance methods, including the '2-level' methods of Watson & Irving (1957) were used. The same procedures were used previously (Doell 1970) and are described in detail there. The value to be used for dispersion due to ancient local anomalies is not known and a value of $1\frac{1}{4}°$ (referred to directions), calculated for Hawaii (Doell & Cox 1963), was used to reduce the between-flow values (subscript B) to ancient field or VGP values (subscript F). This value may be lower than appropriate for Iceland because the geomagnetic field strength, which, in part, induces the anomalies, is less in Hawaii than in Iceland. However, even if the value is doubled it will not change the final values in Table 5 by any meaningful amount.

Palaeosecular variation in Iceland, as expressed by the angular standard deviations and precision parameters given in Table 5, is of little interest until compared with similar data from other localities. There are various ways of making these comparisons, and these have been discussed in detail elsewhere (Cox 1962; Creer 1962; Cox 1970). Detailed reasons for using the angular standard deviation of VGPs about the geographic axis as the most convenient measure of palaeosecular variation for more recent time periods have been given previously (Doell 1970). The main reason for using angular variance of VGPs rather than RM directions is that the contribu-

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Table 5

Dispersion statistics of flow-average data

Data analysed									
Referred to mean VGPs	$\bar{N}$	S_T	S_W	S_B	S_F	K_T	K_W	K_B	K_F
Brunhes-age lava flows	7·96	16·6°	8·3°	16·3°	16·2°	24·1	77·9	25·6	25·9
Matuyama-age lava flows	7·42	18·4	6·5	18·3	18·2	19·8	87·7	23·8	24·1
Pre-Matuyama-age lava flows	6·93	18·4	7·1	18·2	18·1	19·9	79·5	21·2	21·4
Referred to geographic pole									
Brunhes-age lava flows	7·96	16·8°	8·3°	16·5°	16·4°	24·5	77·9	25·5	25·8
Matuyama-age lava flows	7·42	21·1	6·5	21·0	20·9	15·6	87·7	16·0	16·1
Pre-Matuyama-age lava flows	6·93	21·7	7·1	21·5	21·4	14·6	79·5	15·0	15·1
Referred to mean RM direction	$\bar{N}$	S_T	S_W	S_B	S_F	k_T	k_W	k_B	k_F
Brunhes-age lava flows	7·96	9·7°	4·2°	9·6°	9·5°	70·5	303·2	73·2	74·5
Matuyama-age lava flows	7·42	12·3	3·3	12·3	12·2	43·6	333·4	50·6	51·2
Pre-Matuyama-age lava flows	6·93	11·4	3·6	11·3	11·2	51·2	303·1	53·9	54·6
Referred to axial dipole direction									
Brunhes-age lava flows	7·96	9·7°	4·2°	9·6°	9·5°	71·2	303·2	73·4	74·7
Matuyama-age lava flows	7·42	13·9	3·3	13·9	13·8	35·0	333·4	35·5	35·8
Pre-Matuyama-age lava flows	6·93	13·8	3·6	13·7	13·6	35·8	303·1	36·4	36·7

Notes: $\bar{N}$ = Mean number of sample cores per lava.
S, s = Angular standard deviations for VGPs and directions, respectively.
K, k = Precision parameters for VGPs and directions, respectively.
Subscript T = Total dispersion.
Subscript W = Within flow dispersion.
Subscript B = Between flow dispersion (experimental errors removed).
Subscript F = Ancient geomagnetic field dispersion.

tion due to wobble of the main dipole is constant over the entire earth. Thus the difference in values of S_F from area to area is a direct measure of the difference in paleosecular variation due to nondipole field changes. Furthermore, the lowest value of S_F observed anywhere over the entire earth for a given time interval is the maximum contribution due to dipole wobble for the entire earth. The reasons for calculating variances about the rotation pole rather than about VGP mean positions is partly that most paleomagnetic studies (Irving 1964; Cox & Doell 1960) yield VGPs not significantly different from this pole and because rotation effects play a prominent role in geomagnetic theory. Finally, pronounced but very rare excursions of the main dipole to low latitudes (Doell & Cox, in press) cause large contributions to angular variance when they occur in a set of data. For this, and other reasons, it has proven expedient to exclude from analysis all data with VGPs whose latitudes are less than 46° (Doell & Cox, in press). Data with flow-average values of $\alpha_{95} > 9°$ have also been removed from analyses. Thus, the second group of values of S_F in Table 5, for the three different time periods, are the measures of paleosecular variation used here; the other measures are given for the use of those persons preferring them.

With these comments, the angular variances observed in the Icelandic lavas can be compared with those observed elsewhere. For Brunhes age lavas, this has been done in Fig. 9. Also shown are theoretical curves of angular variance versus latitude for a simple dipole wobble, for a simple rotation of the present non-dipole field about the geographic axis, and for a combination of these two (Cox 1970). The value of variance observed at Hawaii, 11°, is taken as the contribution due to dipole wobble since there are reasons for thinking that the non-dipole field has been essentially absent in the Central Pacific during the Brunhes polarity epoch (Doell & Cox 1971, and in press). Ninety-five per cent confidence limits about the mean Icelandic $S_F = 16·4°$ are 19·9° and 13·8° and were calculated using Cox's (1969) methods. As

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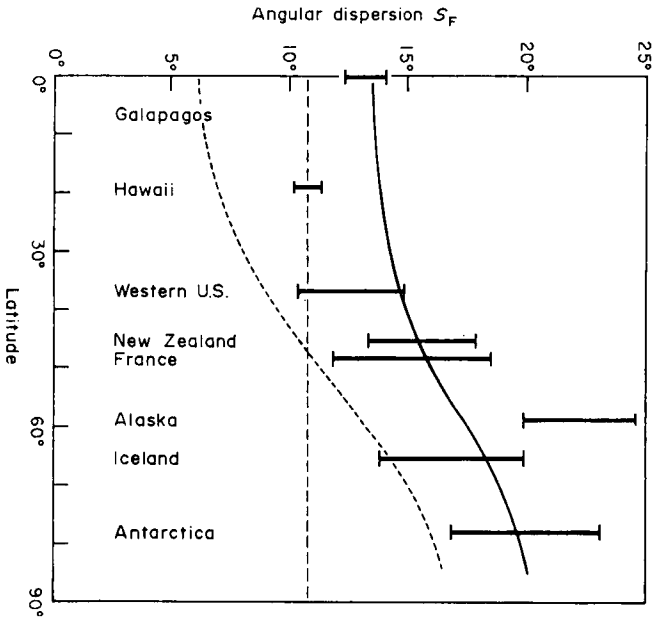


FIG. 9. Brunhes epoch palaeosecular variation, S_F, for several localities plotted against geographic latitude. Vertical bars define 95-per-cent-confidence limits for values of S_F (Cox 1969). The dotted curve is the angular variance expected from rotation of the present non-dipole field, the dashed line that for an 11° wobble of the main dipole, and the solid curve is the variance expected from a combination of the other two (Cox 1970). This figure is adapted from Doell & Cox (1971).

can be seen, these limits include the model of secular variation due to a combination of dipole wobble and a changing non-dipole field with characteristics similar to the present non-dipole field. The limits also include the model without dipole wobble. However, the other available data do not, so that models combining dipole wobble and non-dipole field changes are preferred.

The mean angular variances for the Icelandic Matuyama and pre-Matuyama age lava flows are higher than those for the Brunhes age lavas (Table 5), although the difference is not significant at the 95 per cent confidence level. (The ratio for the Matuyama variance to the Brunhes variance is 1·70, whereas the appropriate F-test ratio is $F_{0.025}$ (27, 27) = 2·16.) In a similar study of lavas from the Massif Central of France (Doell 1969), a variance of 15·2° was found for 31 Brunhes age lavas and a value of 23·2° was obtained from 10 pre-Brunhes age lavas. Again, the older group has a larger palaeosecular variation, but the difference is not significant at a 95 per cent confidence level.

Acknowledgments

It is a pleasure to acknowledge the geological leadership and comradeship provided by Thorleifur Einarsson and David Hopkins during the collection of these samples. Magnetic measurements were carried out by Edward Mankinen at the laboratory of the U.S. Geological Survey, Menlo Park, California, and assistance in the field coring operations was provided by Sveinn Jakobsson and Páll Gústafsson.

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Appendix

Sample location notes (numbers in parentheses are north latitude and east longitude, respectively, of the sampling site followed by the lateral distance spanned by the individual sample cores in meters).

I107: 1961 lava which flowed down NE side of Askja. Sampled along SE edge of flow (65·063°; 343·270°; 46 m).

I106: 1921 lava in Askja Crater. Sampled in outcrops along north edge of lake (65·046°; 343·277°; 52 m).

I105: 1875 lava sampled in northern tongue crossing road between Grimsstadir and Reykjahlid (65·651°; 343·620°; 59 m).

I104: 1783 lava (Skaftáreldahraun). Sampled on east side of Eldvatn at lower bridge Ásar (63·619°; 341·868°; 56 m).

I103: 1729 lava. Sampled at Reykjahlid (65·648°; 343·079°; 113 m).

I102: 2000-year old lava. Sampled in Laxá Canyon below dam (65·816°; 342·681°; 45 m).

I101: 3500-year old lava. Sampled in Laxá Canyon below dam (65·816°; 342·61°; 63 m).

I100: Reykjavik dolerite (64·137°; 338·151°; 22 m).

I99: Lava at Bakakbrúnir (65·696°; 339·698°; 98 m).

I98: Upper of three flows in NE Skagaheidi (Tjörn/Skagi) (66·046°; 339·708°; 123 m).

I97: Middle of three flows in NE Skagaheidi (66·046°; 339·708°; 100 m).

I96: Lower of three flows in NE Skagaheidi (66·046°; 339·708°; 79 m).

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I95: Younger of two lavas at Laxárgljúfur. Two samples from baked zone yielded similar direction (65·816°; 342·681°; 91 m).
I94: Older of two lavas at Laxárgljúfur (65·816°; 342·681°; 80 m).
I93: Younger of three lavas in Jökulsá Canyon just south of Ashöfdi (66·006°; 343·540°; 48 m).
I92: Middle of three lavas in Jökulsá Canyon. Immediately below I93 (66·006°; 343·540°; 46 m).
I91: Older of three lavas in Jökulsá Canyon. Immediately below I92. One sample from baked zone yielded similar direction (66·006°; 343·540°; 35 m).
I90: Lava that forms Ásbyrgi Canyon. Sampled near north end of east side of ' island ' in middle of canyon (66·030°; 343·500°; 107 m).
I89: Younger of two lavas at Dettifoss that lie above a glacial moraine. Unit C of Thorarinsson (1959). Two samples from baked zone yielded similar direction (65·816°; 343·605°; 85 m).
I88: Older of two lavas at Dettifoss that lie below glacial moraine. Unit E of Thorarinsson (1959) (65·816°; 343·605°; 79 m).
I87: Lava at north end of Jökulsá Canyon. Sampled in cliffs along west side near Ashöfdi. This lava is older than I91 (66·025°; 343·559°; 101 m).
I86: Younger of two lavas sampled just SE of farm Svinafell in south Iceland. Sediments and an erosional unconformity separate the two lavas (63·973°; 343·171°; 79 m).
I85: Older of two lavas SE of farm Svinafell (63·973°; 343·171°; 32 m).
I84: Younger of three lavas at Gullfoss. Unit E of Einarsson (1963) (64·327°; 339·874°; 37 m).
I83: Lava immediately below I84. Unit B2 of Einarsson (1963) (64·327°; 339·874°; 59 m).
I82: Lava below I83 with sediments between. Unit B1 of Einarsson (1963) (64·327°; 339·874°; 52 m).
I81: Lava immediately above tillite on Esja Mountain north of Reykjavik. Lowest lava of Grey Basalt Series. One sample from baked zone yielded similar direction (64·228°; 338·312°; 57 m).
I80: Ignimbrite layer at Thorsmörk sampled 100 meters SE of Tourist Cabin (63·684°; 340·516°; 28 m).
I79: Younger of two lavas in Reydará. Youngest lava of sequence on Tjörnes Peninsula (66·075°; 342·690°; 33 m).
I78: Older of two lavas in Reydará (66·075°; 342·690°; 49 m).
I77: Pillow basalt sampled at north end of Húsavik Harbor (66·047°; 342·647°; 43 m).
I76: Lava on surface south of Húsavik (66·027°; 342·659°; 33 m).
I75: Basalt pillows in ocean cliffs south of Húsavik (66·031°; 342·629°; 166 m).
I74: Lava immediately below upper tillite on Grasafjöll (66·067°; 342·860°; 32 m).
I73: Lava within Raudsgjá sediments sampled just south of Raudsgjá (66·190°; 343·008°; 62 m).
I72: Lava above sediments north of Raudsgjá. Three samples from baked zone yielded similar direction. Geologically, could be same lava as I73 (66·190°; 343·008°; 27 m).
I71: Lava immediately below Raudsgjá sediments sampled just northwest of Selbugar, northeast Tjörnes (66·191°; 342·982°; 8 m).
I70: Lava immediately below I71. Sampled in exposures to southeast of I71 (66·195°; 342·978°; 94 m).
I69: Lava immediately below and south-east of I70 (66·198°; 342·975°; 48 m).
I68: Lava immediately below I69, resting on Breidagjá sediments. One sample from baked zone yielded similar direction (66·202°; 342·971°; 34 m).

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I67: Younger of two lavas sampled in sea cliffs just north of Lón, south-east corner of Tjörnes Peninsula (66·125°; 343·041°; 49 m).
I66: Lava immediately below I67 (66·125°; 343·041°; 40 m).
I65: Lava east of Knararbrekkutangi immediately below sediments (66·174°; 342·964°; 99 m).
I64: Lava resting on Breidavik (?) sediments in east Tjörnes coast. Geologically, could be same lava as I65 (66·191°; 342·982°; 63 m).
I63: Lava immediately under Knararbrekkutangi sediments several hundred meters north of Raudsgjá (66·191°; 343·008°; 34 m).
I62: Lava immediately under I63 (66·191°; 343·008°; 70 m).
I61: Lava flow above sediments on east side of Valadalstorfa. Two samples from baked zone yielded similar directions (66·210°; 342·858°; 118 m).
I60: Pillow basalt in Tröllagil 2·5 km south of Breidavik bay coast (66·161°; 342·837°; 70 m).
I59: Lava in Fossgil 1·9 km south of Breidavik bay coast (66·171°; 342·820°; 48 m).
I58: Lava flow in sediments sampled in stream south-east of Sandólar (66·182°; 342·799°; 19 m).
I57: Younger of two lavas in sea cliffs just west of Sandólastöng. Two samples from baked zone yielded similar directions (66·193°; 342·788°; 50 m).
I56: Lava below I57 with 30–40 cm of sediments between. Two samples from baked zone yielded similar directions (66·193°; 342·788°; 74 m).
I55: Lava below 3 m of sediments south-east of Kerling at west end of Breidavik Bay (66·195°; 342·800°; 36 m).
I54: Lava along beach in Breidavik 1 km south-east of Kerling (66·190°; 342·811°; 13 m).
I53: Lava just below I54 exposed to south-east. One sample from baked zone yielded similar direction (66·195°; 342·800°; 38 m).
I52: Younger of two lavas exposed along beach in northern part of Furuvik. Geologically, could be same lava as I53 (66·177°; 342·758°; 32 m).
I39: Lava at southern tip of Hvalvik (66·166°; 342·728°; 33 m).
I38: Lava resting on Tjörnes sediments just north of Litli-Stakkur (66·093°; 342·718°; 60 m).
I37: Younger of two pillow lavas exposed in Skeifá just east of coastal road with sediments above, between, and below (66·128°; 342·740°; 37 m).
I36: Older of two pillow lavas in Skeifá (66·128°; 342·740°; 46 m).
I35: through I28: Eight lavas exposed on the south end of Stöd, north side of Snaefellsnes Peninsula, stratigraphically superposed (I35 youngest) above the Búlandsgil sediments. One baked zone sample below I31 yielded similar direction (64·950°; 336·633°; 38 m (I35), 21 m, 26 m, 14 m, 86 m, 41 m, 29 m, 48 m).
I27: Lava immediately below Búlandsgil sediments on south-west side of Stöd (64·946°; 366·633°; 20 m).
I26: Lava below I27 on south-west Stöd. Also sampled in detail near cross-cutting dike. Only eight samples farthest from dike used in flow-average analysis in Table 3 (64·946°; 336·633°; 71 m).
I25: Lava below I26 on south-west Stöd (64·946°; 336·633°; 61 m).
I24: through I8: Superposed lavas (I24 youngest) exposed in Búlandsgil below road on north edge of Höfdakúlur, north coast of Snaefellsnes Peninsula. One sample from baked zone of I17 yielded a nearly flat direction (64·950°; 336·511°; 20 m (I24), 27 m, 21 m, 14 m, 25 m, 25 m, 22 m, 31 m, 23 m, 17 m, 14 m, 8 m, 3 m, 14 m, 14 m, 9 m, 10 m).
I7: Samples from pillows within Mulberg formation from stream near Máfahlid on south-west Höfdakúlur (64·930°; 336·505°; 43 m).

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I6: Lava within Búlandsgil sediments at Máfahlid. One sample from baked zone yielded a similar direction (64·930°; 336·505°; 91 m).

I5 through I1: Superposed lavas at Máfahlid. I5 is youngest and is immediately below Búlandsgil sediments. One sample each from baked zones of I5, I3, and I2 yielded similar directions to those of the baking flows (64·930°; 336·505°; 65 m (I5), 105 m, 72 m, 55 m, 24 m).

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